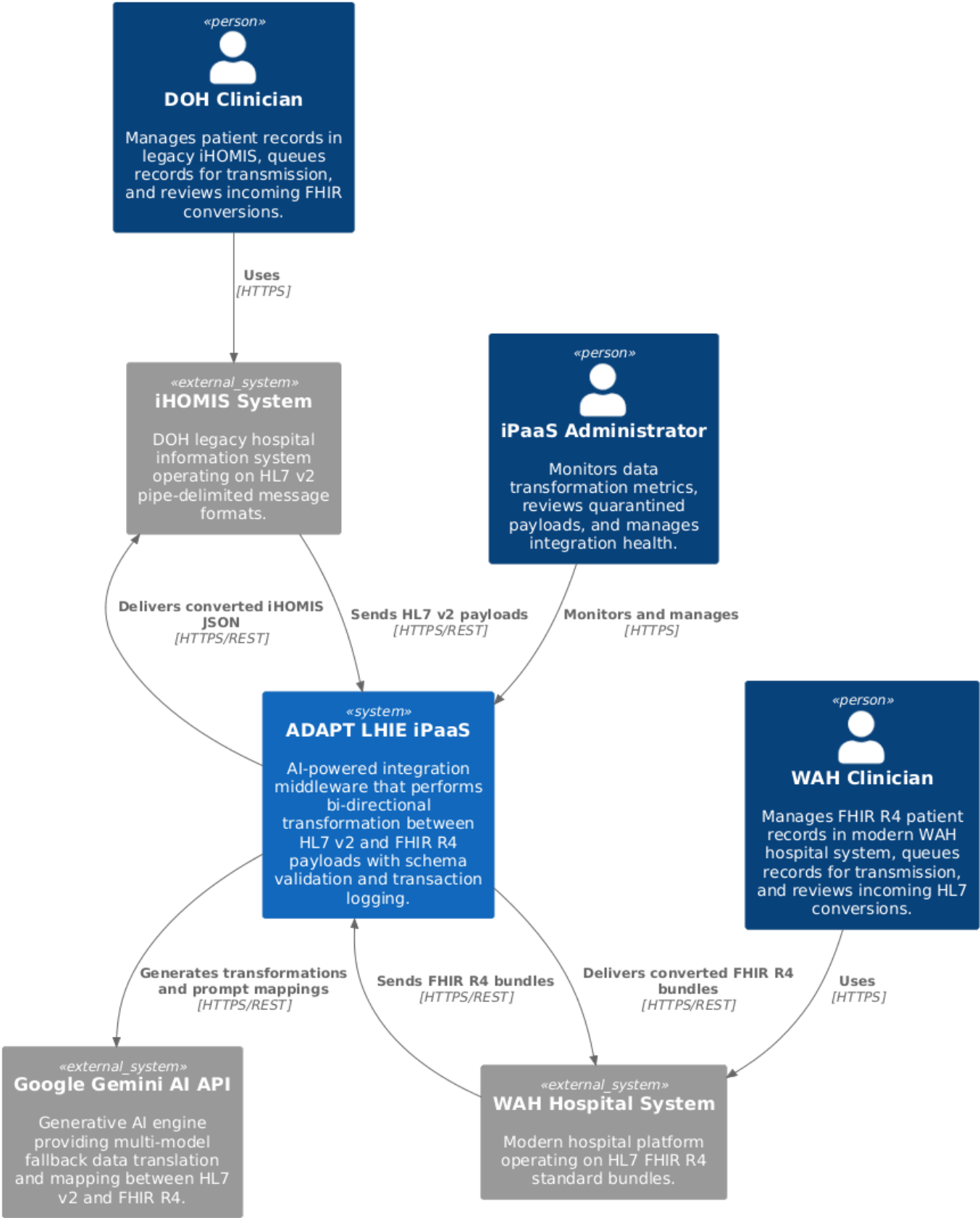


MSYADD

SYSTEM CONTEXT DIAGRAM



```

@startuml ContextDiagram
!include https://raw.githubusercontent.com/plantuml-stdlib/C4-PlantUML/master/C4_Context.puml

Person(dohClinician, "DOH Clinician", "Manages patient records in legacy iHOMIS, queues records for transmission, and reviews incoming FHIR conversions.")

Person(wahClinician, "WAH Clinician", "Manages FHIR R4 patient records in modern WAH hospital system, queues records for transmission, and reviews incoming HL7 conversions.")

Person(admin, "iPaaS Administrator", "Monitors data transformation metrics, reviews quarantined payloads, and manages integration health.")

System(adaptIPaaS, "ADAPT LHIE iPaaS", "AI-powered integration middleware that performs bi-directional transformation between HL7 v2 and FHIR R4 payloads with schema validation and transaction logging.")

System_Ext(ihomisSystem, "iHOMIS System", "DOH legacy hospital information system operating on HL7 v2 pipe-delimited message formats.")

System_Ext(wahSystem, "WAH Hospital System", "Modern hospital platform operating on HL7 FHIR R4 standard bundles.")

System_Ext(geminiApi, "Google Gemini AI API", "Generative AI engine providing multi-model fallback data translation and mapping between HL7 v2 and FHIR R4.")

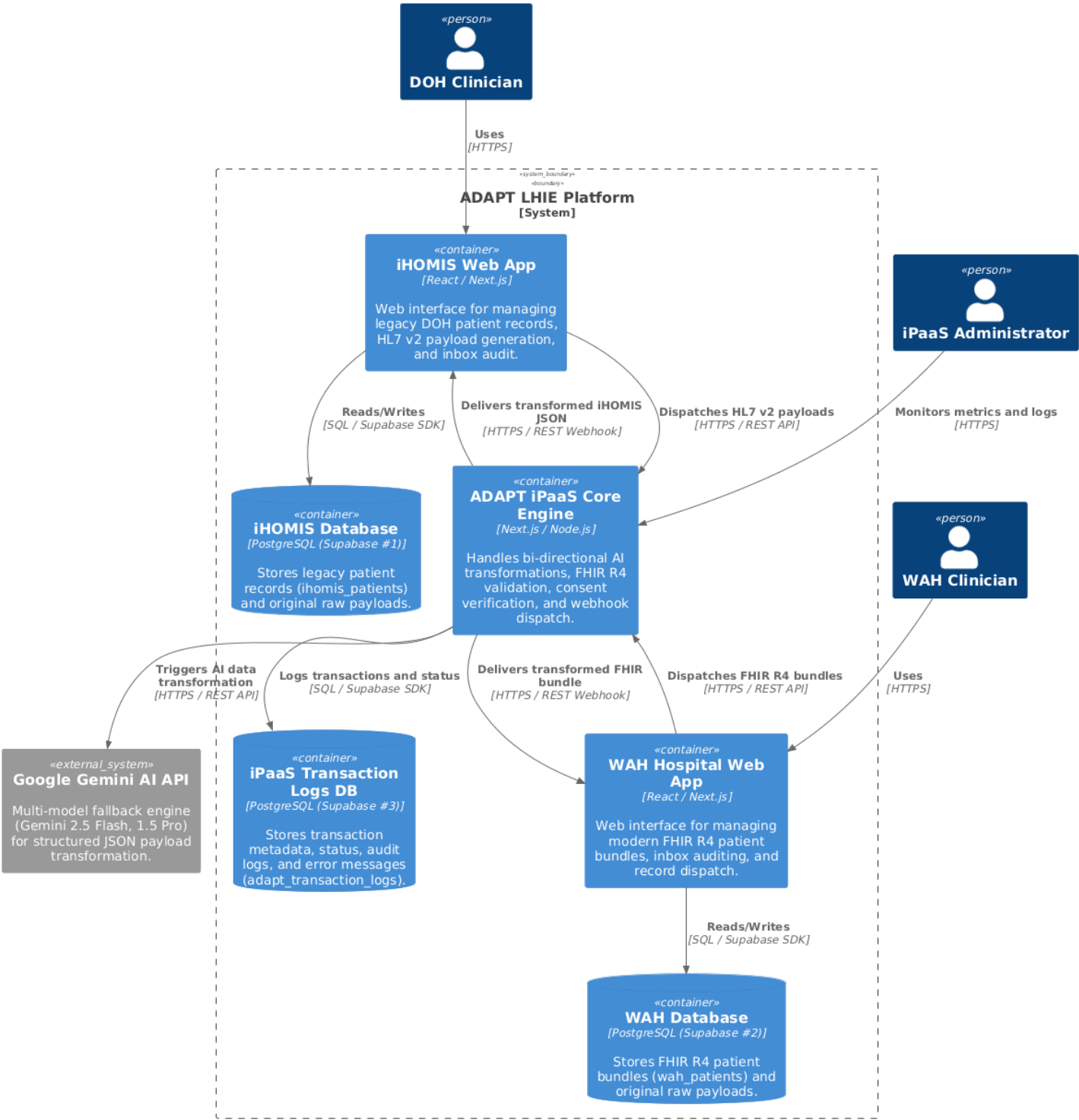
Rel(dohClinician, ihomisSystem, "Uses", "HTTPS")
Rel(wahClinician, wahSystem, "Uses", "HTTPS")
Rel(admin, adaptIPaaS, "Monitors and manages", "HTTPS")

Rel(ihomisSystem, adaptIPaaS, "Sends HL7 v2 payloads", "HTTPS/REST")
Rel(wahSystem, adaptIPaaS, "Sends FHIR R4 bundles", "HTTPS/REST")
Rel(adaptIPaaS, ihomisSystem, "Delivers converted iHOMIS JSON", "HTTPS/REST")
Rel(adaptIPaaS, wahSystem, "Delivers converted FHIR R4 bundles", "HTTPS/REST")

Rel(adaptIPaaS, geminiApi, "Generates transformations and prompt mappings", "HTTPS/REST")
@enduml

```

CONTAINER DIAGRAM



```

@startuml ContainerDiagram
!include https://raw.githubusercontent.com/plantuml-stdlib/C4-PlantUML/master/C4_Container.puml

Person(dohClinician, "DOH Clinician")
Person(wahClinician, "WAH Clinician")
Person(admin, "iPaaS Administrator")

System_Boundary(c1, "ADAPT LHIE Platform") {
    Container(ihomisApp, "iHOMIS Web App", "React / Next.js", "Web interface
for managing legacy DOH patient records, HL7 v2 payload generation,
and inbox audit.")

    Container(wahApp, "WAH Hospital Web App", "React / Next.js", "Web interface
for managing modern FHIR R4 patient bundles, inbox auditing, and record
dispatch.")

    Container(ipaaSApi, "ADAPT iPaaS Core Engine", "Next.js / Node.js",
"Handles bi-directional AI transformations, FHIR R4 validation,
consent verification, and webhook dispatch.")

    ContainerDb(ihomisDb, "iHOMIS Database", "PostgreSQL (Supabase #1)",
"Stores legacy patient records (ihomis_patients) and original raw
payloads.")

    ContainerDb(wahDb, "WAH Database", "PostgreSQL (Supabase #2)", "Stores
FHIR R4 patient bundles (wah_patients) and original raw payloads.")

    ContainerDb(ipaaSDb, "iPaaS Transaction Logs DB", "PostgreSQL
(Supabase #3)", "Stores transaction metadata, status, audit logs, and
error messages (adapt_transaction_logs).")
}

System_Ext(geminiApi, "Google Gemini AI API", "Multi-model fallback engine
(Gemini 2.5 Flash, 1.5 Pro) for structured JSON payload transformation.")

Rel(dohClinician, ihomisApp, "Uses", "HTTPS")
Rel(wahClinician, wahApp, "Uses", "HTTPS")
Rel(admin, ipaaSApi, "Monitors metrics and logs", "HTTPS")

Rel(ihomisApp, ihomisDb, "Reads/Writes", "SQL / Supabase SDK")
Rel(wahApp, wahDb, "Reads/Writes", "SQL / Supabase SDK")

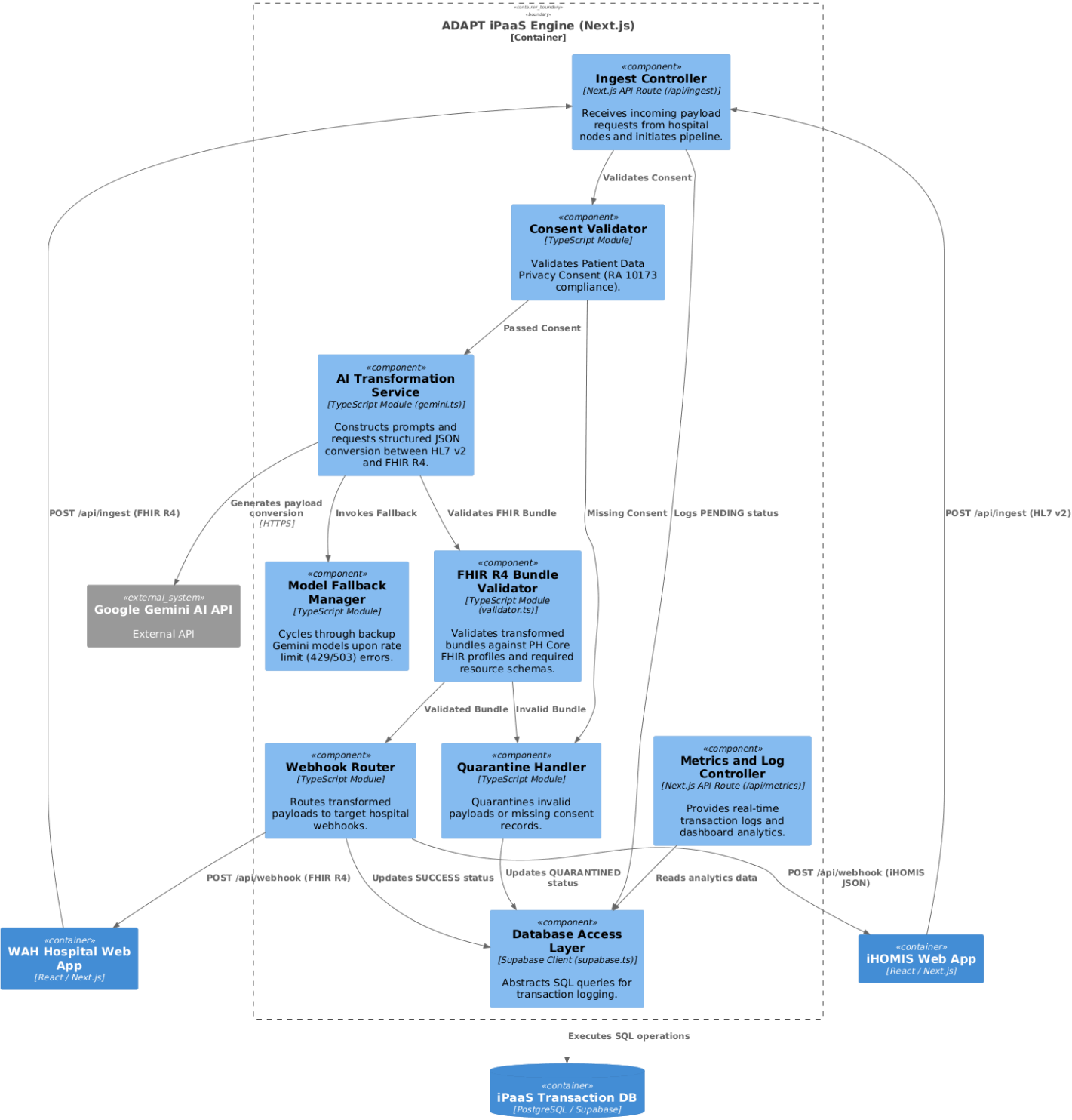
Rel(ihomisApp, ipaaSApi, "Dispatches HL7 v2 payloads", "HTTPS / REST API")
Rel(wahApp, ipaaSApi, "Dispatches FHIR R4 bundles", "HTTPS / REST API")

Rel(ipaaSApi, ipaaSDb, "Logs transactions and status", "SQL / Supabase SDK")
Rel(ipaaSApi, geminiApi, "Triggers AI data transformation", "HTTPS / REST API")

Rel(ipaaSApi, wahApp, "Delivers transformed FHIR bundle", "HTTPS / REST Webhook")
Rel(ipaaSApi, ihomisApp, "Delivers transformed iHOMIS JSON", "HTTPS / REST Webhook")
@enduml

```

COMPONENT DIAGRAM



```

@startuml ComponentDiagram
!include https://raw.githubusercontent.com/plantuml-stdlib/C4-PlantUML/master/C4_Component.puml

Container(ihomisApp, "iHOMIS Web App", "React / Next.js")
Container(wahApp, "WAH Hospital Web App", "React / Next.js")
ContainerDb(database, "iPaaS Transaction DB", "PostgreSQL / Supabase")
System_Ext(geminiApi, "Google Gemini AI API", "External API")

Container_Boundary(api, "ADAPT iPaaS Engine (Next.js)") {
    Component(ingestCtrl, "Ingest Controller", "Next.js API Route (/api/ingest)",
"Receives incoming payload requests from hospital nodes and initiates pipeline.")

    Component(consentSvc, "Consent Validator", "TypeScript Module",
"Validates Patient Data Privacy Consent (RA 10173 compliance).")

    Component(aiTransformSvc, "AI Transformation Service", "TypeScript Module
(gemini.ts)", "Constructs prompts and requests structured JSON conversion
between HL7 v2 and FHIR R4.")

    Component(fallbackMgr, "Model Fallback Manager", "TypeScript Module",
"Cycles through backup Gemini models upon rate limit (429/503) errors.")

    Component(fhirValidator, "FHIR R4 Bundle Validator", "TypeScript Module
(validator.ts)", "Validates transformed bundles against PH Core FHIR profiles
and required resource schemas.")

    Component(webhookRouter, "Webhook Router", "TypeScript Module", "Routes
transformed payloads to target hospital webhooks.")

    Component(quarantineSvc, "Quarantine Handler", "TypeScript Module",
"Quarantines invalid payloads or missing consent records.")

    Component(metricsCtrl, "Metrics and Log Controller", "Next.js API Route
(/api/metrics)", "Provides real-time transaction logs and dashboard analytics.")

    Component(dbAccess, "Database Access Layer", "Supabase Client (supabase.ts)",
"Abstracts SQL queries for transaction logging.")
}

Rel(ihomisApp, ingestCtrl, "POST /api/ingest (HL7 v2)")
Rel(wahApp, ingestCtrl, "POST /api/ingest (FHIR R4)")

Rel(ingestCtrl, consentSvc, "Validates Consent")
Rel(consentSvc, quarantineSvc, "Missing Consent")
Rel(consentSvc, aiTransformSvc, "Passed Consent")

Rel(aiTransformSvc, fallbackMgr, "Invokes Fallback")
Rel(aiTransformSvc, geminiApi, "Generates payload conversion", "HTTPS")

Rel(aiTransformSvc, fhirValidator, "Validates FHIR Bundle")
Rel(fhirValidator, quarantineSvc, "Invalid Bundle")
Rel(fhirValidator, webhookRouter, "Validated Bundle")

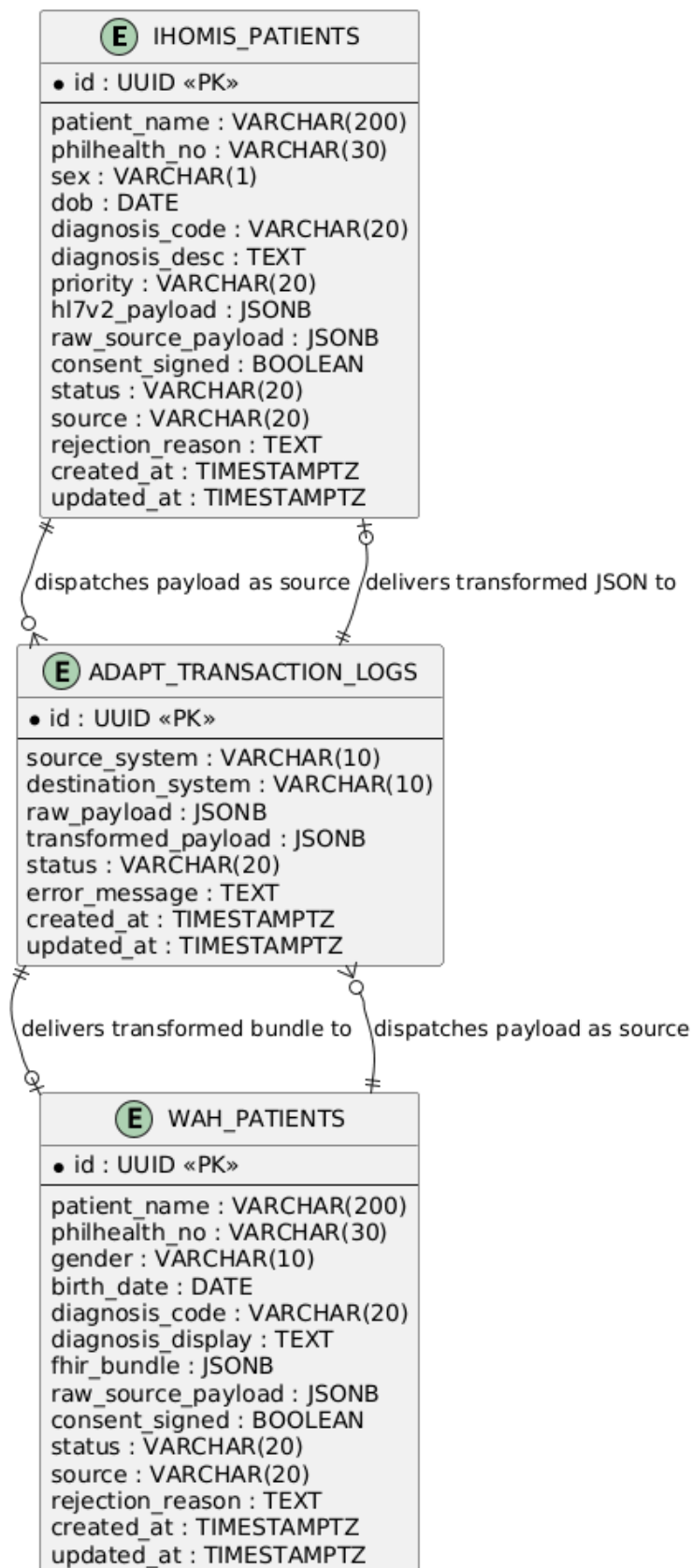
Rel(webhookRouter, wahApp, "POST /api/webhook (FHIR R4)")
Rel(webhookRouter, ihomisApp, "POST /api/webhook (iHOMIS JSON)")

Rel(ingestCtrl, dbAccess, "Logs PENDING status")
Rel(webhookRouter, dbAccess, "Updates SUCCESS status")
Rel(quarantineSvc, dbAccess, "Updates QUARANTINED status")

```

```
Rel(metricsCtrl, dbAccess, "Reads analytics data")
Rel(dbAccess, database, "Executes SQL operations")
@enduml
```

ERD



@startuml ERD

```
entity "IHOMIS_PATIENTS" as ihomis_patients {
    * id : UUID <<PK>>
    --
    patient_name : VARCHAR(200)
    philhealth_no : VARCHAR(30)
    sex : VARCHAR(1)
    dob : DATE
    diagnosis_code : VARCHAR(20)
    diagnosis_desc : TEXT
    priority : VARCHAR(20)
    hl7v2_payload : JSONB
    raw_source_payload : JSONB
    consent_signed : BOOLEAN
    status : VARCHAR(20)
    source : VARCHAR(20)
    rejection_reason : TEXT
    created_at : TIMESTAMPTZ
    updated_at : TIMESTAMPTZ
}
```

```
entity "WAH_PATIENTS" as wah_patients {
    * id : UUID <<PK>>
    --
    patient_name : VARCHAR(200)
    philhealth_no : VARCHAR(30)
    gender : VARCHAR(10)
    birth_date : DATE
    diagnosis_code : VARCHAR(20)
    diagnosis_display : TEXT
    fhir_bundle : JSONB
    raw_source_payload : JSONB
    consent_signed : BOOLEAN
    status : VARCHAR(20)
    source : VARCHAR(20)
    rejection_reason : TEXT
    created_at : TIMESTAMPTZ
    updated_at : TIMESTAMPTZ
}
```

```
entity "ADAPT_TRANSACTION_LOGS" as adapt_transaction_logs {
    * id : UUID <<PK>>
    --
    source_system : VARCHAR(10)
    destination_system : VARCHAR(10)
    raw_payload : JSONB
    transformed_payload : JSONB
    status : VARCHAR(20)
    error_message : TEXT
    created_at : TIMESTAMPTZ
    updated_at : TIMESTAMPTZ
}
```

```
ihomis_patients ||--o{ adapt_transaction_logs : "dispatches payload as source"
wah_patients ||--o{ adapt_transaction_logs : "dispatches payload as source"
adapt_transaction_logs ||--o| wah_patients : "delivers transformed bundle to"
adapt_transaction_logs ||--o| ihomis_patients : "delivers transformed JSON to"
@enduml
```